

Fixed Point Structure of Gradient Flow Exact Renormalization Group

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Based on arXiv:2201.04111 with Y.Abe(Kyoto) and Y.Hamada(KEK)

Summary of today's talk

- We study the fixed point structure of “Gradient Flow Exact Renormalization Group” (GFERG), a new framework to define the Wilson action via a gradient flow equation
- We consider the GFERG equation associated with a general gradient flow equation for scalar field theories
- We show that the fixed points are the same as those of the Wilson–Polchinski equation in general
- Furthermore, we discuss that the GFERG equation has a similar RG flow structure around a fixed point to the WP equation

Contents

- Introduction (3)
- Review of GFERG (10)
- Fixed point structure of GFERG for scalar field theories (14)
- Conclusion and Future directions (3)

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Exact Renormalization Group (ERG)

- A framework to study physics under **varying the energy scale**
- The only way to define the continuum limit of QFTs
- **The Wilson action S_τ** is intuitively defined by **integrating out higher momentum modes** of the fields:

$$e^{-S_\tau} := \int D\phi_{p>\Lambda} e^{-S_0}$$

($\Lambda := \Lambda_0 e^{-\tau}$, Λ_0 : cutoff)

- τ -dependence of the Wilson action S_τ is described by **a differential equation** $\rightarrow$ **“ERG equation”**

Wilson-Polchinski equation

- A typical example of the ERG equation:

$$\partial_\tau S_\tau = \int_p \left\{ \left[\left(2p^2 + \frac{D+2-\eta_\tau}{2} \right) + p_\mu \frac{\partial}{\partial p_\mu} \right] \phi_i(p) \frac{\delta S_\tau}{\delta \phi_i(p)} + \left(2p^2 + 1 - \frac{\eta_\tau}{2} \right) \left(\frac{\delta^2 S_\tau}{\delta \phi_i(p) \delta \phi_i(-p)} - \frac{\delta S_\tau}{\delta \phi_i(p)} \frac{\delta S_\tau}{\delta \phi_i(-p)} \right) \right\}$$

$$\partial_\tau \text{ (circle } S_\tau) = \text{ (circle } S_\tau) + \text{ (circle } S_\tau \text{ with self-loop)} - \text{ (circle } S_\tau \text{ --- circle } S_\tau)$$

- This equation defines a renormalization procedure non-perturbatively

(η_τ : anomalous dimension)

[J. Polchinski *Nucl. Phys. B* 231 (1984) 269-295]

(We work on the dimensionless framework and D -dimensional Euclidean space)

Gauge invariance in ERG

- The Wilson action with the naive UV cutoff is inconsistent with gauge invariance
- The gauge transformation mixes higher and lower momentum modes:

$$A_\mu^a(p) \rightarrow A_\mu^a(p) - p_\mu \omega^a(p) - i f^{abc} \int_q \omega^b(p - q) A^c(q)$$

- Can we define a Wilson action in a manifestly gauge invariant manner?

Renormalization and Diffusion

- The solution to the WP equation can be written in the following form:

$$e^{S_\tau[\phi]} = \hat{s}_\phi^{-1} \int D\phi' \prod_{x,i} \delta(\phi_i(x) - e^{\tau(D-2)/2} Z^{1/2} \varphi'_i(t, x e^\tau)) \hat{s}_{\phi'} e^{S_{\tau=0}[\phi']}$$

$$(\hat{s}_\phi := \exp\left(-\int_x \frac{\delta^2}{\delta\phi_i(x)^2}\right) : \text{“scrambler”})$$

- φ' is a solution to the diffusion equation:

$$\partial_t \varphi'(t, x) = \partial_x^2 \varphi'(t, x), \quad \varphi'(0, x) = \phi'(x)$$

where $t := e^{2\tau} - 1$

- This representation implies that the coarse-graining by the diffusion can be used to define a RG flow

Gradient Flow (GF)

- This one-parameter deformation of fields via the diffusion equation has been studied in the context of “gradient flow”
- The gradient flow is a method to construct composite operators without the equal-point singularity
- Correlation functions are UV finite with wave function renormalization:

$$Z_t^{-n/2} \langle \varphi(t_1, x_1) \varphi(t_2, x_2) \cdots \varphi(t_n, x_n) \rangle_\phi < \infty$$

even for the equal point case (e.g. $x_1 = x_2$)

[F. Capponi, L. Debbio, S. Ehret, R. Pellegrini, A. Rago 1512.02851]

Gradient Flow for gauge fields

- The gradient flow equation for gauge fields

[M. Lüscher, P. Weisz 1101.0963]

$$\partial_t B'_\mu = \underbrace{D'_\nu G'_{\nu\mu}}_{\text{“diffusion”}} + \alpha_0 \underbrace{D'_\mu \partial_\nu B'_\nu}_{\text{“gauge fixing”}}$$

with $B'_\mu(0, x) = A'_\mu(x)$

- $D'_\nu G'_{\nu\mu}$ can be expanded as

$$D'_\nu G'_{\nu\mu} = \partial_x^2 B'_\mu + \dots$$

the gradient flow equation is regarded as a
“gauge-covariant” diffusion equation

Gradient Flow ERG (GFERG)

- S_τ for gauge fields can be defined via the GF equation

[H. Sonoda and H. Suzuki 2012.03568]

$$e^{-S_\tau[A_\mu^a]} := \hat{s}_A^{-1} \int [DA_\mu'^a] \prod_{x', a, \mu} \delta \left(A_\mu^a(x) - e^{\tau(D-4)} B_\mu'^a(t, x' e^\tau) \right) \hat{s}_{A'} e^{-S_{\tau=0}[A_\mu'^a]}$$

$$(\hat{s}_A := \exp \left[-\frac{1}{2} \int_x \frac{\delta^2}{\delta A_\mu^a(x) \delta A_\mu^a(x)} \right]): \text{“scrambler” for gauge fields})$$

- This S_τ is invariant under gauge transformation!
- The way to define the Wilson action via the gradient flow equation is called
“Gradient Flow Exact Renormalization Group”

GFERG equation

- We get the counterpart of the WP equation by differentiating S_τ with respect to τ :

$$\partial_\tau e^{-S_\tau} =$$

$$\int_x \frac{\delta}{\delta A_\mu^a(x)} \left[-2D_\nu F_{\nu\mu}^a - 2\alpha_0 D_\mu \partial_\nu A_\nu^a - \left(\frac{D-2+\eta_\tau}{2} + x^\mu \frac{\partial}{\partial x^\mu} \right) A_\mu^a \right]_{A \rightarrow A + \delta/\delta A} e^{-S_\tau}$$

- This equation is called “the GFERG equation”
- It includes higher-functional derivatives more than two.
(The WP equation includes up to a second derivative)

Recent Studies

- Inclusion of matter fermions, the chiral symmetry and the axial anomaly

[Y.Miyakawa and H.Suzuki 2106.11142]

[Y.Miyakawa 2201.08181]

- Gauge fixing, ghosts and perturbative analysis around the Gaussian fixed point in QED

[Y.Miyakawa, H.Sonoda and H.Suzuki 2111.15529]

- GFERG equation for the LPI effective action and gauge symmetry in QED

[H.Sonoda and H.Suzuki 2201.04448]

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Our motivation

- The appropriate gradient flow eq. highly depends on details of the theory, such as its symmetry and interactions
- The form of GFERG eq. depends on that of the GF eq., and then takes a different form for each theory
- This means that the GFERG eq. for general scalar field theories is no longer given by the WP eq.
- Is GFERG consistent with the conventional ERG?

Our motivation

- The gradient flow equation for the $O(N)$ non-linear sigma model

$$\partial_t \varphi_i = \partial_x^2 \varphi_i - (\varphi_j \partial_x^2 \varphi_j) \varphi_i$$

[H. Makino, H. Suzuki 1410.7538]

- But this model and the linear sigma model in three dimension belong to the same universality class
- “Does GFERG give the same prediction as the conventional ERG for the IR behavior of a theory?”
- These facts strongly motivate us to study the fixed points and the critical exponents in GFERG

Our work

- We study the fixed points of the general GFERG eq. for scalar fields
 - The fixed points of the WP eq. appear in the $\tau \rightarrow \infty$ limit along the GFERG flow
- We show the GFERG eq. has a similar RG flow structure around a fixed point
 - Scaling dimensions of relevant or marginal operators are the same, while those of irrelevant operators can be different
- GFERG gives the same prediction as the conventional ERG for the IR behavior of a theory

General gradient flow equation

- General form of gradient flow equations

$$\partial_\tau \varphi_i = \underbrace{\partial_x^2 \varphi_i}_{\text{WP part}} + \sum_{n=n_{\min}}^{\infty} \int_{x_1, \dots, x_n} \underbrace{f_i^{i_1, \dots, i_n}(x; x_1, \dots, x_n; \partial_{x_1}, \dots, \partial_{x_n}) \varphi_{i_1}(\tau, x_1) \cdots \varphi_{i_n}(\tau, x_n)}_{\text{extra terms}}$$

- The Wilson action can be defined in the same way as the linear sigma model:

$$e^{S_\tau[\phi]} = \hat{s}_\phi^{-1} \int D\phi' \prod_x \delta(\phi_i(x) - e^{\tau(D-2)/2} Z_\tau^{1/2} \varphi'_i(t, x e^\tau)) \hat{s}_{\phi'} e^{S_{\tau=0}[\phi']}$$

General GFERG equation

- GFERG equation

$$\partial_\tau e^{-S_\tau[\Phi]} = (\text{WP part})$$

$$+ \sum_{n=n_{\min}}^{\infty} \lambda^{n-1}(\tau) \int_{x, x_1, \dots, x_n} \frac{\delta}{\delta \phi_i(x)} \left\{ f_i^{i_1, \dots, i_n} \left(\phi_{i_1}(x_1) + \int_{y_1} \mathcal{D}(x_1 - y_1) \frac{\delta}{\delta \phi_{i_1}(y_1)} \right) \times \dots \right. \\ \left. \times \left(\phi_{i_n}(x_n) + \int_{y_n} \mathcal{D}(x_n - y_n) \frac{\delta}{\delta \phi_{i_n}(y_n)} \right) \right\} e^{-S_\tau[\Phi]}$$

where $\lambda(\tau) := e^{-\tau(D-2)/2} Z_\tau^{-1/2}$

- It contains the extra non-linear terms and higher-functional derivatives

Fixed point action

- The fixed point action S^* is defined as

$$\partial_\tau S^* = 0 \quad \text{and} \quad S^* := \lim_{\tau \rightarrow \infty} S_\tau$$

- S^* satisfies

$$0 = (\text{WP part}) + \sum_{n=n_{\min}}^{\infty} \lambda^{n-1}(\infty) \int_{x, x_1, \dots, x_n} \frac{\delta}{\delta \phi_i(x)} \left\{ f_i^{i_1, \dots, i_n} \left(\phi_{i_1}(x_1) + \int_{y_1} \mathcal{D}(x_1 - y_1) \frac{\delta}{\delta \phi_{i_1}(y_1)} \right) \times \dots \right. \\ \left. \times \left(\phi_{i_n}(x_n) + \int_{y_n} \mathcal{D}(x_n - y_n) \frac{\delta}{\delta \phi_{i_n}(y_n)} \right) \right\} e^{-S^*[\Phi]}$$

- Asymptotic behavior of $\lambda(\tau)$

$$\lambda(\tau) \sim \exp(-\tau(D - 2 + \eta)/2) \quad \frac{\eta}{2} := \frac{d}{d\tau} \log Z_\tau \Big|_{\tau=\infty}$$

- $\lambda(\infty)$ should vanish ($\rightarrow$ Next slide)

Vanishing λ

- The connected two point function at the fixed point theory

$$\langle \phi(x)\phi(0) \rangle_{\text{connected}} \propto \frac{1}{x^{D-2+\eta}}$$

- It should become zero when $|x| \rightarrow \infty$
 - $\rightarrow D - 2 + \eta > 0$ is required
 - $\rightarrow \lambda(\infty)$ ($\lambda(\tau) \sim e^{-\tau(D-2+\eta)/2}$) and the extra terms vanish and only the WP part survives
 - $\Rightarrow S^*$ satisfies the fixed point condition of the WP eq.
- The fixed points of the WP eq. appears in the $\tau \rightarrow \infty$ limit along the GFERG flow

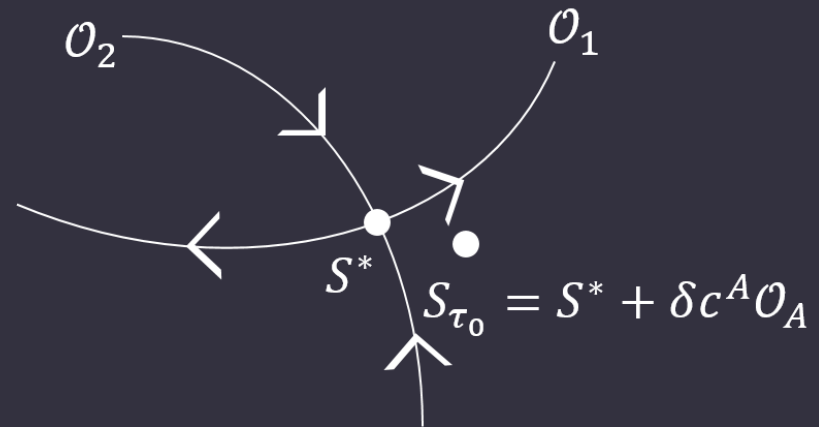
RG flow around fixed point

- Let us study the τ -dependence of S_τ after a long time $\tau = \tau_0 \gg 1$ so that $\lambda(\tau_0) \sim e^{-\tau_0(D-2+\eta)/2} \ll 1$
- Consider perturbing S_τ from a fixed point S^* at $\tau = \tau_0$ as

$$S_{\tau_0} = S^* + \sum_A \delta c^A \mathcal{O}_A$$

$$|\delta c^A| \ll 1$$

$\mathcal{O}_A$: a complete set of operators
(defined in the next slide)



Solution in the $\tau \rightarrow \infty$ limit

- We have two small quantities: δc^A , $\lambda_0 := e^{-\tau_0(D-2+\eta)/2}$
- Solution to the GFERG eq. up to the leading order

$$S_\tau = S^* + \sum_A (\delta c^A e^{x_A \tau'} - \lambda_0^{n_{\min}-1} (e^{-(n_{\min}-1)(D-2+\eta)\tau'/2} - e^{x_A \tau'}) h^A) \mathcal{O}_A$$

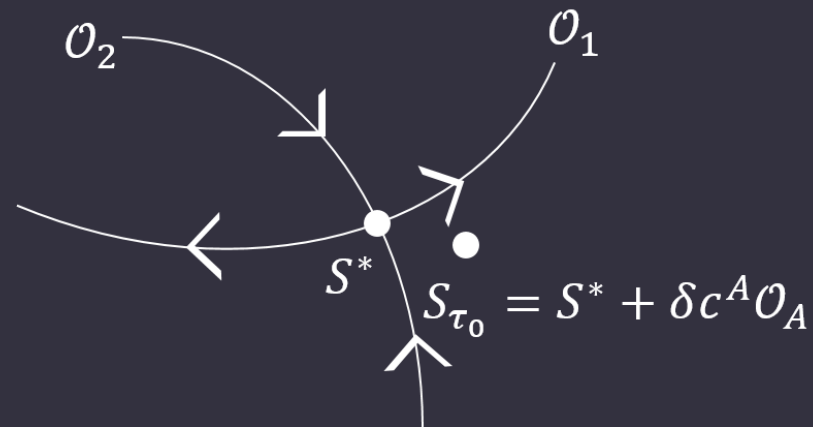
where $\tau = \tau_0 + \tau'$

$\mathcal{O}_A$: eigenoperators of the linearized WP eq. around S^*

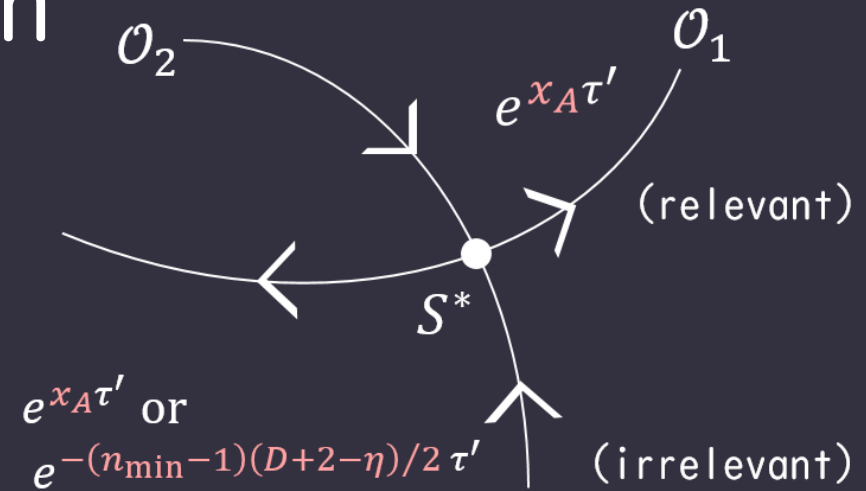
x_A : eigenvalue of $\mathcal{O}_A$

$n_{\min}$: minimum order of the non-linear terms in the GF eq.

h^A : some constant



Scaling dimension



- Scaling Dimension d_A

$$d_A = \max(x_A, -(n_m - 1)(D - 2 + \eta)/2)$$

$$x_A \geq 0 \text{ (relevant or marginal)} \Rightarrow d_A = x_A$$

$$x_A < 0 \text{ (irrelevant)} \Rightarrow d_A = -\min(|x_A|, (n_m - 1)(D - 2 + \eta)/2)$$

- d_A of relevant or marginal operators are x_A
 $\Rightarrow$ critical exponents are the same as those of the WP eq.

$O(N)$ non-linear sigma model

- Let us illustrate our results with the $O(N)$ non-linear sigma model in $4 - \epsilon$ dimensions
- Lagrangian

$$\mathcal{L} = \frac{1}{2g} (\partial_\mu \phi_i)^2, \quad (\phi_i)^2 = 1$$

- Its IR physics is characterized by **Wilson-Fisher fixed point** in $4 - \epsilon$ dimensions

$$S_{\text{WF}}^* = \int_x \frac{1}{2} (\partial_\mu \phi_i)^2 + \frac{m_*^2}{2} \phi_i^2 + \frac{\lambda_*}{8} (\phi_i^2)^2 + O(\epsilon^2)$$

$$m_*^2 := -\frac{\epsilon N + 2}{4N + 8}, \quad \lambda^* := \epsilon \frac{8\pi^2}{N + 8}$$

[K.Wilson, M.Fisher *Phys.Rev.Lett.* 28 (1972) 240-243]

[S.Dutta, B.Sathiapalan, H.Sonoda 2003.02773]

GFERG of $O(N)$ NL sigma model

- Gradient flow equation

$$\partial_\tau \varphi_i = \partial_x^2 \varphi_i - (\varphi_j \partial_x^2 \varphi_j) \varphi_i$$

[H. Makino, H. Suzuki 1410.7538]

- GFERG equation

$$\partial_\tau e^{-S_\tau[\Phi]} = (\text{WP part})$$

$$\begin{aligned} & -\lambda^2(\tau) \int_x \frac{\delta}{\delta \phi_i(x)} \left(\left(\phi_j(x) + \int_y \mathcal{D}(x-y) \frac{\delta}{\delta \phi_j(y)} \right) \right. \\ & \times \partial_x^2 \left(\phi_j(x) + \int_y \mathcal{D}(x-y) \frac{\delta}{\delta \phi_j(y)} \right) \left(\phi_i(x) + \int_y \mathcal{D}(x-y) \frac{\delta}{\delta \phi_i(y)} \right) \Bigg) e^{-S_\tau[\Phi]} \end{aligned}$$

Fixed point, Scaling dimension

- Anomalous dimension at WF fixed point

$$\frac{\eta}{2} = \frac{N+2}{(N+8)^2} \frac{\epsilon^2}{2} + O(\epsilon^3)$$

$$\rightarrow \lambda(\tau) \sim \exp\left(-\frac{\tau}{2}\left(2 - \epsilon + \frac{N+2}{(N+8)^2} \epsilon^2\right)\right) \rightarrow 0 \quad (\tau \rightarrow \infty)$$

- Scaling dimensions

$$- \mathcal{O}_1 = (\phi_i^2)/2 \rightarrow d_1 = \max(2 - \epsilon, -(D - 2 + \eta)) = 2 - \epsilon > 0$$

$$(\text{critical exponent } \nu^{-1} := d_1 = 2 - \epsilon)$$

$$- \mathcal{O}_2 = (\phi_i^2)/2 - (\phi_i^2)^2/8 \rightarrow d_2 = \max(-\epsilon, -(D - 2 + \eta)) = -\epsilon < 0$$

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Conclusion

- Studied the fixed point structure of the GFERG equation associated with a **general** gradient flow equation for **scalar field theories**
- Showed that **the fixed points are the same** as those of the Wilson–Polchinski equation in general
- Discussed that the GFERG equation has **a similar RG flow structure** around a fixed point to the WP equation
- GFERG gives the **same** prediction as the conventional ERG for the IR behavior of a theory

Future Direction

- (Non-abelian) gauge theory
- Gravity and asymptotic safety
“Wilson action with manifest diffeomorphism invariance”
- Scalar field theories with a non-trivial target space in two dimensions
e.g.) $O(N)$ NL sigma model, CP^{N-1} model
- Effects of topological terms in QFT

Thank you!

Back up

GFERG is ERG?

- GFERG is **NOT** considered to be a kind of ERG but an alternative framework to **study the low-energy physics in the Wilsonian sense**
- Reasons:
 1. The scaling dimensions are **different** from those of the WP equation
(Different **renormalization schemes** should provide **the same scaling dimensions** within ERG by a field redefinition)
 2. The effective couplings do **not follow the scaling law** around fixed points
(They should scales as $\lambda^A \rightarrow e^{x_A \tau} \lambda^A$ under the coarse-graining of $x \rightarrow xe^\tau$)

Loophole

- Our arguments do not cover the case where $\lambda(\tau)$ and then the GFERG equation become τ -independent

- η_τ satisfies

$$\eta_\tau = 2 - D$$

- Examples: the pure Yang-Mills theory with $D = 4$,
the $O(N)$ non-linear sigma model with $D = 2$
(Note: the counterpart of $\lambda(\tau)$ in the YM theory is $e^{-\tau(D-4)/2} Z_\tau^{1/2}$)
- The GFERG equation can be regarded as a ERG equation in this case

GF eq. of $O(N)$ NL sigma

- Gradient flow equation

$$\partial_t \varphi_i = \partial_x^2 \varphi_i - (\varphi_j \partial_x^2 \varphi_j) \varphi_i$$

- It obviously preserves the $O(N)$ symmetry ($\varphi_i \rightarrow O_{ij} \varphi_j$)
- The constraint of the fields ($\varphi_i^2 = 1$) is respected, i.e., if $\varphi_i^2(0, x) = \phi_i^2(x) = 1$, $\varphi_i^2(t, x) = 1$ for $\forall t$
- The correlation functions of φ_i are UV finite without any additional wave function renormalization in two dimensions
- If we solve the constraint ($\phi_N = \pm(1 - \sum_a^{N-1} \phi_a^2)^{\frac{1}{2}}$), the appropriate gradient flow equation is

$$\partial_t \varphi_a = \partial_x^2 \varphi_a + \left(\partial_\mu \varphi_b \partial_\mu \varphi_b + \frac{(\varphi_b \partial_\mu \varphi_b)^2}{1 - \varphi_c^2} \right) \varphi_a$$

General non-linear sigma model

- Lagrangian

$$\mathcal{L} = \frac{1}{2} G^{nm}(\phi) \partial_\mu \phi_n \partial^\mu \phi_m$$

- The Wilson action of GFERG is **not covariant** under target space diffeomorphism of the fields

$$e^{S_\tau[\phi]} = \hat{s}_\phi^{-1} \int D\phi' \prod_x \delta(\phi_i(x) - e^{\tau(D-2)/2} Z_\tau^{1/2} \varphi'_i(t, x e^\tau)) \hat{s}_{\phi'} e^{S_{\tau=0}[\phi']}$$

$$\hat{s}_\phi := \exp\left(-\int_x \frac{\delta^2}{\delta\phi_i(x)^2}\right)$$

- This expression should be interpreted as one for the models **embedded in higher-dimensional Euclidean space** with **some constraints** for the fields
- The appropriate GF equation is required to preserve those constraints instead

Constraints

- Consider a theory whose fields are constrained as $F[\phi] = 0$
e.g.) the $O(N)$ NL sigma model: $F = \phi_i^2 - 1$
- S_τ preserves constraints of the fields in the sense that $F[\lambda(\tau)\phi] = 0$ holds in the modified correlation function:

$$\ll F[\lambda(\tau)\phi] \phi_{i_1} \cdots \phi_{i_n} \gg_{S_\tau} = 0$$

(Recall:

$$e^{S_\tau[\phi]} = \hat{s}_\phi^{-1} \int D\phi' \prod_x \delta(\phi_i(x) - e^{\tau(D-2)/2} Z_\tau^{1/2} \varphi'_i(t, x e^\tau)) \hat{s}_{\phi'} e^{S_{\tau=0}[\phi']} \quad)$$

Scaling theorem

- S_τ of GFERG satisfies the scaling theorem:

[H. Sonoda, H. Suzuki 1901.05169]

$$\begin{aligned} & \ll \phi_{i_1}(p_1 e^\tau) \cdots \phi_{i_n}(p_n e^\tau) \gg_{S_\tau} \\ &= e^{-\tau n(D+2)/2} Z_\tau^{n/2} \prod_{i=1}^n \frac{K(p_i e^\tau)}{K(p_i)} \ll \phi_{i_1}(p_1) \cdots \phi_{i_n}(p_n) \gg_{S_{\tau=0}} \end{aligned}$$

$\ll \mathcal{O} \gg_S := \int D\phi \hat{s}_\phi \mathcal{O} e^{S[\phi]}$: the modified correlation function

$K(p_i) := e^{-p^2}$: the cutoff function

- The solution to the WP equation holds this property
- S_τ describes the response of the system to the scale transformation (the coarse-graining)

RG invariance of Z

- Partition function

$$Z := \int D\phi e^{S_{\tau=0}[\phi]}$$

- The GFERG flow **preserves** the partition function:

$$\int D\phi e^{S_\tau} = \int D\phi e^{S_{\tau=0}} = Z$$

- We call S_τ of GFERG the “Wilson” action in the sense that it holds this property and satisfies the scaling theorem

Details of Solution to GFERG eq.

- Solution to the GFERG equation in the $\tau \rightarrow \infty$ limit

$$S_\tau = S^* + \sum_A (\delta c^A e^{x_A \tau'} - \lambda_0^{n_{\min}-1} (e^{-(n_{\min}-1)(D-2+\eta)\tau'/2} - e^{x_A \tau'}) h^A) \mathcal{O}_A$$

$$h^A = \frac{g^A}{x_A + D - 2 + \eta}$$

$$\sum_A g^A \mathcal{O}_A := e^{S^*} \int_{x, x_1, \dots, x_n} \frac{\delta}{\delta \phi_i(x)} \left\{ f_a^{i_1, \dots, i_{n_{\min}}} \left(\phi_{i_1}(x_1) + \int_{y_1} \mathcal{D}(x_1 - y_1) \frac{\delta}{\delta \phi_{i_1}(y_1)} \right)^{n_{\min}} e^{-S^*} \right.$$

Inclusion of matter fermions

[Y.Miyakawa and H.Suzuki 2106.11142]

- Wilson action for gauge fields and matter fermions

$$e^{S_\tau[A,\psi,\bar{\psi}]} := \hat{s}_A^{-1} s_\psi^{-1} \int DA' D\psi' D\bar{\psi}' \prod_{x'', \nu, b} \delta \left(A_\nu^b(x'') - e^\tau g_\tau^{-1} B_\nu'^b(t, x'' e^\tau) \right) \\ \times \delta \left(\psi(x'') - e^{\tau(D-1)/2} Z_\tau^{1/2} \chi'(t, x'' e^\tau) \right) \delta \left(\bar{\psi}(x'') - e^{\tau(D-1)/2} Z_\tau^{1/2} \bar{\chi}'(t, x'' e^\tau) \right) \\ \times \hat{s}_{A'} \hat{s}_{\psi'} e^{S_{\tau=0}[A', \psi', \bar{\psi}']}$$

$$\hat{s}_\psi := \exp \left(- \int_x \frac{\delta_L}{\delta \psi(x)} \frac{\delta_L}{\delta \bar{\psi}(x)} \right): \text{scrambler for fermions}$$

- $\chi'(t, x)$ is the solution to the following GF equation:

$$\partial_t \chi'(t, x) = [D_\mu D_\mu - \alpha_0 \partial_\mu B_\mu'^a(t, x) T^a] \chi'(t, x), \quad \chi'(0, x) = \psi'(x)$$

Chiral symmetry

- S_τ is **NOT** invariant under the chiral transformation

$$\psi(x) \rightarrow (1 + i\alpha\gamma_5)\psi(x)$$

due to the scrambler $\hat{s}_\psi := \exp\left(-\int_x \frac{\delta_L}{\delta\psi(x)} \frac{\delta_L}{\delta\bar{\psi}(x)}\right)$

- It is invariant under the **modified** chiral symmetry generated by

$$\hat{\Gamma}_5 := \hat{s}_\psi^{-1} \hat{\gamma}_5 \hat{s}_\psi,$$

where $\hat{\gamma}_5$ is the generator of the ordinary one:

$$\hat{\gamma}_5 := \int_x \gamma_5 \psi(x) \frac{\delta_L}{\delta\psi(x)} + \bar{\psi}(x) \gamma_5 \frac{\delta_L}{\delta\bar{\psi}(x)}$$

Modified chiral symmetry

- $\hat{\Gamma}_5$ acts on S_τ as

$$e^{-S_\tau} \hat{\Gamma}_5 e^{S_\tau} = \int_x \frac{\delta_R S_\tau}{\delta \psi} \gamma_5 \psi + \bar{\psi} \gamma_5 \frac{\delta_L S_\tau}{\delta \bar{\psi}} - 2 \frac{\delta_R S_\tau}{\delta \psi} \gamma_5 \frac{\delta_L S_\tau}{\delta \bar{\psi}} + 2 \text{tr} \left[\gamma_5 \frac{\delta_L}{\delta \psi} \frac{\delta_R S_\tau}{\delta \bar{\psi}} \right]$$

- When S_τ is the **bilinear** in the fermion field as

$$S_\tau = - \int_x \bar{\psi}(x) D \psi(x) \dots,$$

the modified chiral symmetry ($\hat{\Gamma}_5 e^{S_\tau} = 0$) implies the **Ginsparg-Wilson relation**:

$$\gamma_5 D + D \gamma_5 + 2 D \gamma_5 D = 0$$

(Recall that we work on the dimensionless framework)

Gauge fixing in QED

[Y.Miyakawa, H.Sonoda and H.Suzuki 2111.15529]

- When perform a perturbative calculation, we should consider **the gauge fixing** and **the ghosts**
- If we introduce the ghosts $\bar{c}(x), c(x)$ and suppose they **decouples** from A_μ and ψ , the GFERG equation requires

$$S_{\text{gh}} = \int_x \bar{c}(x) \frac{\partial^2}{E(-e^{-2\tau}\partial^2) - \partial^2} c(x)$$

where $S_\tau = S[A, \psi] + S_{\text{gh}}$

- The Ward-Takahashi identity requires

$$S_\tau^{\text{inv}} := S_\tau + \frac{1}{2} \int_x \frac{1}{\xi E(-e^{-2\tau}\partial^2) e^{2\partial^2} - \partial^2} \left(\partial_\mu A_\mu(x) \right)^2$$

should be invariant the **modified** gauge transformation

ξ : the gauge fixing parameter

$E(x)$: an arbitrary function analytic at $x=0$

Modified gauge invariance

- S_{τ}^{inv} is required to be invariant under the modified gauge transformation:

$$\delta A_{\mu}(x) = \frac{\xi E(-e^{-2\tau} \partial^2) e^{2\partial^2} - \partial^2}{\xi E(-e^{-2\tau} \partial^2) e^{2\partial^2}} \partial_{\mu} \omega(x),$$

$$\delta \psi(x) = ie\omega(x)\psi(x),$$

$$\delta \bar{\psi}(x) = -ie\omega(x)\bar{\psi}(x)$$

- S_{τ} is required to transform under it as

$$\delta S_{\tau} = \int_x A_{\mu}(x) \frac{\partial^2}{\xi E(-e^{-2\tau} \partial^2) e^{2\partial^2}} \partial_{\mu} \omega(x)$$

IPI effective action

[H. Sonoda and H. Suzuki 2201.04448]

- The generating functional $\Gamma[\mathcal{A}_\mu, \Psi, \bar{\Psi}]$ for **one-particle irreducible diagrams** is defined as

$$\Gamma_\tau[\mathcal{A}_\mu, \Psi, \bar{\Psi}] := S_\tau[A_\mu, \psi, \bar{\psi}] + \int_x \left[\frac{1}{2} (A_\mu - \mathcal{A}_\mu)^2 + i(\bar{\psi} - \bar{\Psi})(\psi - \Psi) \right],$$

where A_μ , ψ and $\bar{\psi}$ are set to be the solutions to

$$\mathcal{A}_\mu = A_\mu + \frac{\delta S_\tau}{\delta A_\mu}, \Psi = \psi + i \frac{\delta_L S_\tau}{\delta \psi}, \bar{\Psi} = \bar{\psi} + i \frac{\delta_R S_\tau}{\delta \bar{\psi}}$$

- Γ_τ is invariant under the **ordinary** gauge transformation, while S_τ^{inv} is invariant under the **modified** one

Material

